

Learning a Dynamic Cross-Modal Network for Multispectral Pedestrian Detection

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ABSTRACT

Multispectral pedestrian detection that enables continuous (day and night) localization of pedestrians has numerous applications. Existing approaches typically aggregate multispectral features by a simple element-wise operation. However, such a local feature aggregation scheme ignores the rich non-local contextual information. Further, we argue that a local tight correspondence across modalities is desired for multi-modal feature aggregation. To address these issues, we introduce a multispectral pedestrian detection framework that comprises a novel dynamic cross-modal network (DCMNet), which strives to adaptively utilize the local and non-local complementary information between multi-modal features. The proposed DCMNet consists of a local and a non-local feature aggregation module. The local module employs dynamically learned convolutions to capture local relevant information across modalities. On the other hand, the non-local module captures non-local cross-modal information by first projecting features from both modalities into the latent space and then obtaining dynamic latent feature nodes for feature aggregation.

Comprehensive experiments are performed on two challenging benchmarks: KAIST and LLVIP. Experiments reveal the benefits of the proposed DCMNet, leading to consistently improved detection performance on diverse detection paradigms and backbones. When using the same backbone, our proposed detector achieves absolute gains of 1.74% and 1.90% over the baseline Cascade RCNN on the KAIST and LLVIP datasets.

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CCS CONCEPTS

• **Computing methodologies** → **Scene understanding; Object detection.**

KEYWORDS

pedestrian detection, multi-modal fusion, dynamic learning

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1 INTRODUCTION

Pedestrian detection is an important computer vision task that strives to detect pedestrians in an image. Modern pedestrian detection methods generally take RGB images as input and perform well in an adequately illuminated environment. However, their performance largely depends on the appropriate illumination, hampering their performance under low-light scenarios. This issue can be mainly attributed to the inability of RGB cameras in acquiring useful information about pedestrians under low-light scenarios. Compared with RGB cameras, infrared cameras are sensitive to temperature changes and capture infrared images that provide more information in nighttime. However, the infrared images captured by infrared cameras usually suffer from insufficient details, low resolution, etc., which are also essential for accurate detection. To this end, the multispectral pedestrian detection methods [11] aiming at combining the advantages of the RGB images and infrared images have become one of the most effective ways to solve the problem of pedestrian detection at various levels of illumination.

Multispectral pedestrian detection has gained popularity [45, 51, 59] due to its applicability in diverse set of applications ranging from self-driving cars to video surveillance. Existing multispectral pedestrian detection methods generally rely on the fusion of features from RGB and infrared input images. Liu *et al.* [23] studied the effectiveness of fusing low-level, mid-level, and high-level features

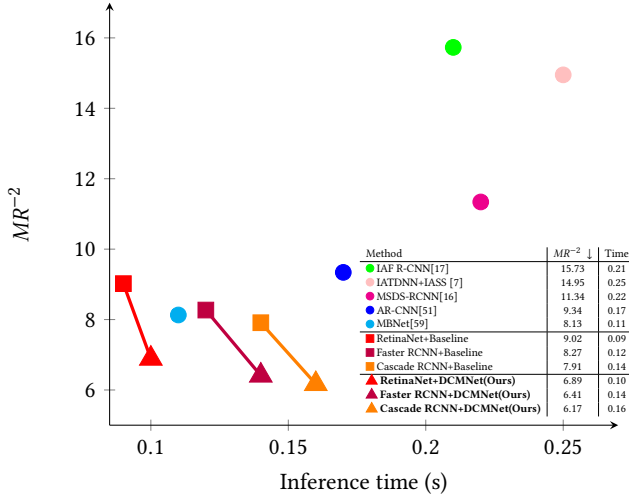


Figure 1: Miss rate (%) vs. speed (s) comparisons on the KAIST test set using a single TitanX GPU. Our DCMNet is generic and achieves the state-of-the-art performance while being integrated into diverse detection frameworks.

and classification scores. The study shows that the best detection performance can be obtained by fusing middle-level convolutional features. Recently, MBNet [59] introduces a modality aware fusion to mitigate the imbalance between the color and thermal modalities. In order to accurately align features from two different modalities, Zhang *et al.* [51] introduced a method that addresses the position shift between two modalities. The aforementioned feature fusion based methods generally employ element-wise fusion of local features from two different modalities.

In this work we distinguish two key properties that are desired while fusing features from different modalities to achieve accurate multispectral pedestrian detection. (i) *Capturing complementary information from different modalities*: As discussed before, RGB images are often useful in highly-illuminated scenarios while infrared images are particularly useful to capture pedestrian details under low-light scenarios. Here, the information from the RGB and infrared modalities is not equally important. Hence, for accurate detection, it is desired to obtain local tight corresponding between two modalities by identifying and fusing important complementary features from both modalities. (ii) *Exploiting both local and non-local information from different modalities for the feature fusion*: Most of existing multispectral pedestrian detection methods perform local feature aggregation from different modalities by utilizing operations such as element-wise addition, concatenation, *etc.*. However, such local feature aggregations often fail to capture the long-range dependencies. In order to simultaneously achieve these desired properties, we introduce a novel dynamic cross-modal network (DCMNet) that aggregates local and global complementary information from RGB and infrared modalities.

Contributions: The main contributions of this paper are:

- We propose an effective feature aggregation network, termed dynamic cross-modal network (DCMNet), which comprises a local dynamic cross-modal (LDCM) module and a non-local dynamic cross-modal (NDCM) module. To reduce the noises during fusion, Our LDCM module employs dynamically learned convolutions

to capture relevant and local complementary information from two modalities for effective feature fusion. To enrich the non-local contextual information between two modalities, our NDCM captures long-range dependencies across different modalities by aggregating scale-specific non-local information in dynamically learned latent feature spaces.

- We perform experiments on two multispectral pedestrian detection benchmarks: LLVIP [12] and KAIST [11]. On both datasets, our DCMNet provides superior performance. On the challenging KAIST dataset, the integration of our DCMNet into a RetinaNet based detector provides higher accuracy (lower miss rate) and speed compared to the state-of-the-art methods [59] (see Fig. 1).
- Our DCMNet is generic and provides consistent improvements in the detection accuracy while being integrated into diverse detectors (including single-stage, two-stage, and anchor-free detectors), and various backbones (including CNN and Transformer).

2 RELATED WORK

Object Detection and Pedestrian Detection: Object detection aims at locating and classifying objects in images, which is the basis for several other computer vision tasks (*e.g.*, action recognition [13, 14], tracking [47, 48], and video recognition [33]). Recently, object detection approaches based on deep learning have exhibited state-of-the-art performance, which can be roughly classified into anchor-based methods [1, 21, 24, 28, 29, 32, 56] and anchor-free methods [2, 36]. Anchor-based methods can be divided into two categories: single-stage methods [21, 24, 56] and two-stage methods [1, 32]. Single-stage methods directly regress object boxes from the predefined anchors and classify them. RetinaNet [21] is a classical single-stage method that solves the class-imbalance problem by introducing a focal loss. Two-stage methods first generate proposals from the predefined anchors and then classify and regress these proposals. Faster R-CNN [32] is a popular two-stage object detection [1] and pedestrian detection frameworks [27, 52, 54]. Cascade R-CNN [1] consists of a sequence of detectors that are trained with increasing IoU thresholds for better detection accuracy. Pedestrian detection is a closely related task that aims at localizing and classifying pedestrians in RGB input images. It has received a considerable interest in the research community due to its relevance in various surveillance and robotic applications. Similar to object detection, pedestrian detection can be roughly classified into anchor-based pedestrian detection methods [5, 10, 25, 30, 40, 41, 55, 57, 58] and anchor-free pedestrian detection methods [18, 19, 26, 35].

Multispectral Pedestrian Detection: The aforementioned pedestrian detection methods generally take RGB input images. However, RGB cameras are ineffective at acquiring valuable information about pedestrians at night. Hence, single modality pedestrian detection methods are generally not robust to illumination variations. Multispectral pedestrian detection, which adopts multiple sensors of different types, has good robustness to the changes of illumination. It has become important applications for self-driving cars and video surveillance, where the RGB image and infrared images can provide complementary visual information. Multispectral pedestrian detection approaches based on deep learning have exhibited state-of-the-art performance [16, 23, 45, 50, 59]. Liu *et al.* [23] exploited the

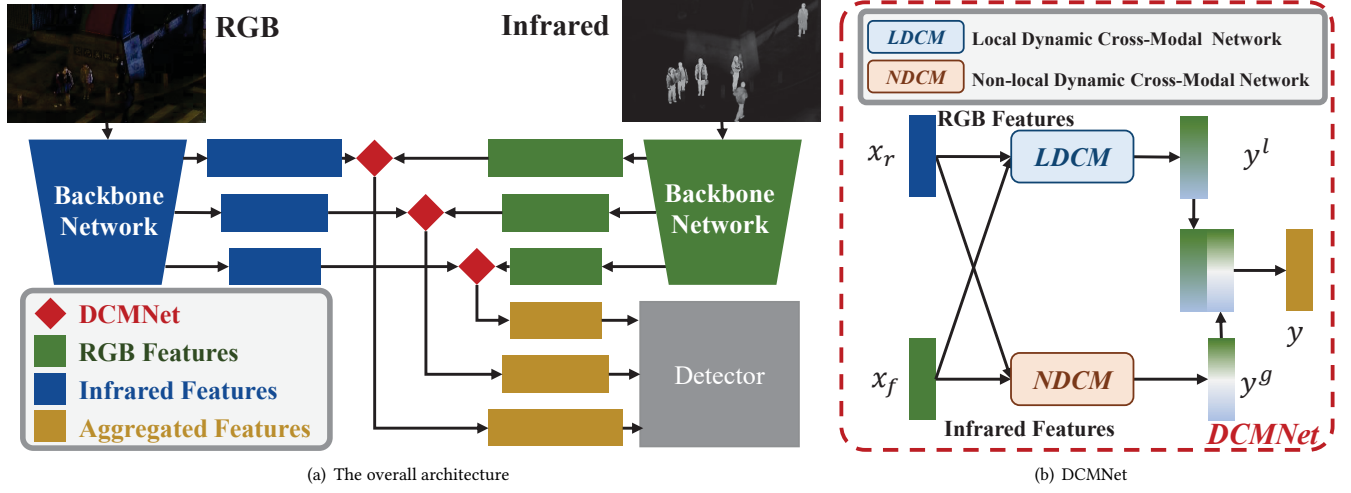


Figure 2: (a) The overall architecture of the proposed multispectral pedestrian detection framework. It comprises a two-stream backbone network taking RGB and infrared images as inputs, a novel dynamic cross-modal network (DCMNet), and a detector for pedestrian classification and localization. The focus of our design is DCMNet for effective aggregation of RGB and infrared features. The DCMNet takes backbone features from the RGB and infrared streams as its inputs and performs scale-specific aggregation of these features. (b) The DCMNet comprises a local dynamic cross-modal (LDCM) module and a non-local dynamic cross-modal (NDCM) module to respectively capture the local and non-local complementary information across two modalities.

effectiveness of four feature fusion techniques at different stages, in which a halfway fusion model that fuses middle-level convolutional features obtains the best detection performance. MSDS-RCNN [16] improves the pedestrian detection performance by jointing pedestrian detection and segmentation. AR-CNN [51] introduces a RFA module to shift and align the features from color images and infrared images. MBNet [59] utilizes a differential modality aware fusion module and an illumination aware feature alignment module to solve the modality imbalance problem. GAFF [45] learns inter- and intra-modality attention to fuse multispectral features.

Limitations: Most of the existing methods aggregate features from different modalities by naive element-wise operations (e.g., element-wise addition, concatenation, etc.). These methods generally treat the two modalities as equal, therefore, such naive element-wise feature aggregation operations may lead to capturing irrelevant and distracting information from different modalities. And some methods employ weighted addition by using the predicted illumination value. Note that an error in illumination prediction gives rise to indiscriminative features. In addition, these methods often fail to model the long-range feature dependencies which are also important to reason the existence and locations of pedestrians. To address these issues, we propose a dynamic cross-modal network (DCMNet) comprising a local dynamic cross-modal (LDCM) module and a non-local dynamic cross-modal (NDCM) module. The LDCM module employs dynamically learned convolution kernels to extract the discriminative features by reducing the heterogeneity between two modalities. Our NDCM module further improves the detection accuracy by capturing the cross-modality long-range dependencies. Different to [39, 53] that capture long-range dependencies in single modality (RGB) images, our NDCM module captures complementary long-range dependencies across different modalities for improved multispectral pedestrian detection.

3 METHOD

Overall Architecture: Fig. 2(a) shows the overall architecture of the proposed multispectral pedestrian detection method. It consists of a two-stream backbone network (including feature pyramid networks (i.e., FPN [20]). For brevity, FPN is omitted in subsequent sections.) taking RGB and infrared input images, a dynamic cross-modal network (DCMNet), and a detector. The focus of our design is our novel DCMNet to effectively aggregate multispectral features at different scales for accurate pedestrian detection. At first, the two-stream backbone network extracts multi-scale features from the RGB and infrared images. Then our DCMNet (see Fig. 2(b)) aggregates the RGB and infrared features at different scales by capturing the local and non-local complementary information between the two modalities. Our DCMNet comprises of a LDCM module and a NDCM module to capture local and non-local complementary information between the modalities, respectively. Finally, the aggregated features from DCMNet are utilized in the detector for the classification and localization of pedestrians.

3.1 Dynamic Cross-Modal network (DCMNet)

We propose a DCMNet to aggregate multispectral features. The DCMNet contains two modules: local dynamic cross-modal (LDCM) module and non-local dynamic cross-modal (NDCM) module. The LDCM module employs dynamically learned convolutions to reduce the heterogeneity between the two modalities by exploring the local complementary information. The NDCM module enforces features with long-range correspondences of the cross modalities to help reason the existence and location of pedestrians. The LDCM and the NDCM modules aggregate the multispectral features effectively in local and global aspects and obtain enriched multispectral features for achieving accurate detection results. In the following, we describe the LDCM and NDCM modules in detail.

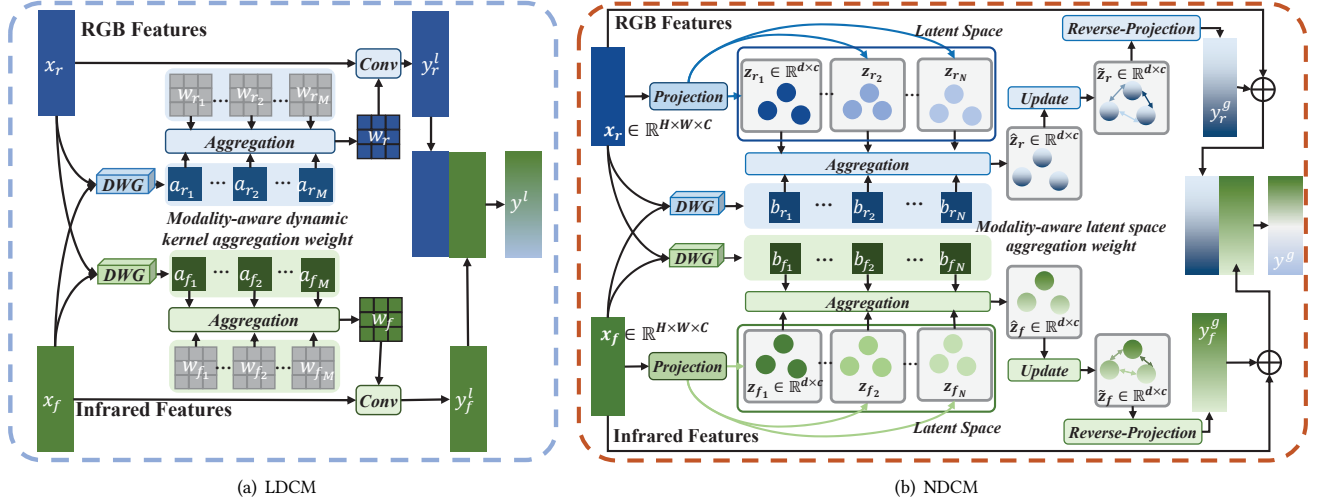


Figure 3: Structures of our LDCM and NDCM modules. (a) The LDCM module captures local cross-modal information by employing modality-aware dynamic convolutional kernels. The dynamic convolutional kernels are generated by the dynamic kernel aggregation weights from a dynamic weight generation (DWG) module (shown in Fig. 4). (b) The NDCM module first projects the RGB and infrared features into several latent spaces, and then employs the aggregation weights predicted from the DWG to aggregate the latent spaces, at last captures cross-modal long-range dependencies in the aggregated latent spaces.

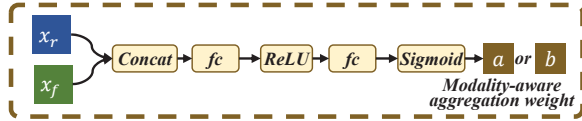


Figure 4: The structure of the DWG module.

3.2 LDCM Module

In this subsection, we review the standard convolution and then detail our LDCM module.

Standard Convolution. The input and output of the convolution operations can be defined as x and y , respectively. The convolution operations can be defined as:

$$y = w^T x + b, \quad (1)$$

$x \in \mathbb{R}^{h \times w \times c_{in}}$ and $y \in \mathbb{R}^{h \times w \times c_{out}}$ represent the input and output of the convolution operations. $w \in \mathbb{R}^{K \times K \times c_{in} \times c_{out}}$ and $b \in \mathbb{R}^{c_{out}}$ represent the weight and bias of convolution operations. h and w represent the height and width of the input and output features (Here, we assume that the height and width of the input and output features are the same.). c_{in} and c_{out} represent the channel numbers of input and output features, respectively. It can be observed that the convolutional kernel of the standard convolution is independent of the image content. Therefore, the standard convolution is unable to explore the complementary information across two modalities. Recently, some data-driven convolution operations are proposed [4, 8, 42]. Though these methods achieve success in RGB-based computer vision tasks, these methods can not extract the information between two modalities. In this paper, we propose a local feature aggregation module, termed as LDCM module. LDCM module employs dynamically learned convolutional kernels

to make the local correspondences tighter by exploring the local complementary information.

LDCM. The detailed structure of the local dynamic cross-modal (LDCM) module is shown in Fig.3(a). The input of LDCM module is the RGB features and the infrared features which are extracted by the two-stream backbone network from RGB images and infrared images, respectively. We define the RGB features as x_r , and the infrared features as x_f . In order to extract local complementary information, we employ dynamically learned convolutional kernels depending on both RGB features and infrared features.

For simplicity, we only illustrate the RGB branch of the LDCM module, which can be defined as:

$$y_r^l = w_r^T x_r, \quad (2)$$

where y_r^l is the enriched features for the RGB branch. The bias is not employed in our method, and the RGB-aware dynamically learned convolutional kernels w_r are used to learn the useful information between the two modalities. The RGB-aware dynamically learned convolutional kernels are generated by employing the RGB-aware dynamic kernel aggregation weight to aggregate several convolutional kernels. Specifically, the RGB-aware dynamically learned convolutional kernels w_r can be computed as follows:

$$w_r = \sum_{m=1}^M w_{r_m} \cdot a_{r_m}, \quad (3)$$

where a_{r_m} is the the RGB-aware dynamic kernel aggregation weight, $w_{r_m} \in \mathbb{R}^{K \times K \times c \times c}$, $m = 1, 2, \dots, M$ is a group of convolutional kernels, M is the number of convolutional kernels to be aggregated, K is the convolutional kernel size, c is the channel number of the RGB features x_r .

Next, we detail the dynamic weight generation (DWG) module (shown in Fig. 4) which is used to obtain RGB-aware dynamic

kernel aggregation weight a_{r_m} . Specifically, We first combine the RGB features x_r and the infrared features x_f through an operator of concatenation as:

$$x_c = x_r || x_f, \quad (4)$$

where x_c are the combined features. $||$ represents the concatenation operator. And then we employ the combined features to compute the RGB-aware dynamic kernel aggregation weight a_r as follows:

$$a_r = \sigma(\mathbf{FC}_b(\mathbf{ReLU}(\mathbf{FC}_a(\text{GAP}(x_c)))) \in \mathbb{R}^M, \quad (5)$$

where $\mathbf{FC}_a$ and $\mathbf{FC}_b$ represent fully-connected layers, and the output dimension of these two fully-connected layers are $\frac{c}{2}$ and M , respectively. c is the channel number of RGB features x_r , GAP represents global average pooling, and σ represents the sigmoid function. It can be observed that the dynamically learned convolutional kernels jointly depend on both of the RGB features and infrared features, which would help our LDCM module explore the local complementary information to reduce irrelevant and distracting information during the fusion.

Similar to the RGB branch, we can compute the features for the infrared branch y_f^l . The local enriched features y^l are obtained by combining the y_r^l and y_f^l through concatenation as:

$$y^l = \mathbf{Conv}(y_r^l || y_f^l), \quad (6)$$

where $\mathbf{Conv}$ is a convolution operation that is used to reduce the channel numbers, and y^l is the output of the LDCM module.

3.3 NDCM Module

Existing non-local operations can only capture long-range dependencies of a single modality, which is unable to model the long-range dependencies across the two modalities. It will limit the effects of the long-range dependencies on reasoning the location and existence of pedestrians. To address this issue, we propose an efficient non-local dynamic cross-modal (NDCM) module. More specifically, we first project the features from the grid spaces into latent spaces to obtain latent feature nodes, where one latent feature node represents a region of features in the grid space. We then aggregate these latent spaces by utilizing dynamic aggregation weights. Next, we utilize the relations among different latent feature nodes to model the relations between the dynamic latent feature nodes to capture the long-range dependencies. At last, we reverse-project the features from the latent space to the grid space and obtain features with enriched cross-modality non-local information. Fig. 3(b) shows the detailed structure of the NDCM module. For simplicity, we only detail the RGB branch of the NDCM module in the following.

We first use different projection metrics to project the features from the grid space into a group of latent spaces. The features in the latent space, termed as latent feature nodes, can be computed as follows:

$$z_{r_n} = \mathbf{P}_{r_n} x_{r_n} \in \mathbb{R}^{d \times c}, n = 1, 2, \dots, N. \quad (7)$$

where N is the number of latent spaces. n is the index of latent spaces, for the n -th latent space, z_{r_n} is the generated RGB-aware latent feature nodes, $\mathbf{P}_{r_n}$ is the feature projection matrix learned by the RGB features x_r , d is the number of the latent nodes, c is the number of channels.

Then the DWG module is employed to learn RGB-aware dynamic latent space aggregation weight b_{r_n} from the cross-modal combined feature x_c , which is utilized in the subsequent aggregation of latent feature nodes. The detailed computation as follow:

$$b_{r_n} = \sigma(\mathbf{FC}_b(\mathbf{ReLU}(\mathbf{FC}_a(\text{GAP}(x_c)))) \in \mathbb{R}^N. \quad (8)$$

where x_c are the concatenated features generated by concatenating the RGB features x_r and the infrared features x_f output from the backbone network. $\mathbf{FC}_a$ and $\mathbf{FC}_b$ denote the fully-connected layers, the output dimensions of which are $\frac{c}{2}$ and N , respectively. c is the channel number of RGB features x_r . The generated RGB-aware dynamic latent space aggregation weight b_r are used to aggregate latent feature nodes dynamically.

Next, the RGB-aware dynamic latent space aggregation weight b_{r_n} is employed to aggregate the different latent spaces as follows:

$$\hat{z}_r = \sum_{n=1}^N z_{r_n} \cdot b_{r_n} \in \mathbb{R}^{d \times c}. \quad (9)$$

Then we construct a fully-connected adjacency matrix $\mathbf{A}_r$ to update the latent feature nodes as follows:

$$\begin{aligned} \tilde{z}_r &= \mathbf{A}_r z_r \\ &= \mathbf{Normalize}(\hat{z}_r \hat{z}_r^T) \hat{z}_r, \end{aligned} \quad (10)$$

where $\mathbf{Normalize}$ denotes the L2 normalization. Finally, we reverse-project the latent feature node to the grid features as follow:

$$y_r^g = x_r + \mathbf{R}_r \tilde{z}_r. \quad (11)$$

where $\mathbf{R}_r$ is the reverse-projection matrix which is learned by the concatenated features x_c .

Similarly to the computation process of the enriched RGB features y_r^g , we can compute the enriched infrared features y_f^g . The non-local cross-modality features y^g are obtained by combining the y_r^g and y_f^g with concatenation as:

$$y^g = \mathbf{Conv}(y_r^g || y_f^g), \quad (12)$$

where $\mathbf{Conv}$ is the convolution operation that is utilized to reduce the channel numbers, and y^g is the output of the NDCM module. **Combining LDCM and NDCM.** Finally, we concatenate the features y^l and y^g as follows:

$$y = \mathbf{Conv}(y^l || y^g). \quad (13)$$

The output features y are fed into the detector to predict the confidence scores and bounding-box coordinates of pedestrians.

4 EXPERIMENTS

4.1 Datasets

LLVIP is a challenging dataset for multispectral pedestrian detection including 12,025 training images and 3,463 test images. We use the COCO-style Average Precision AP [22] as evaluation metrics. **KAIST** is a popular multispectral pedestrian detection dataset. The models are trained on the annotations provided by [51] and [16], and tested on the improved test annotations [23]. We report performance using standard average-log miss rate, which is computed over the false positive per image (FPPI) range of $[10^{-2}, 10^0]$ [6]. We select MR^{-2} as evaluation metrics, for which lower values indicate better detection performance.

Table 1: Analyzing the impacts of different modules on the LLVIP *test* set. All the methods use the same training data, input scale and experimental settings.

Methods	AP $\uparrow$
Baseline (Concatenation)	56.9
Baseline + LDCM	58.2
Baseline + NDCM	58.3
Our DCMNet: Baseline+LDCM+NDCM	58.9

4.2 Implementation Details

A popular object detection toolbox MMDetection¹ [3] based on PyTorch [31] is used to implement our method. For both datasets, our method is trained on one NVIDIA GeForce RTX 3090 GPU. The batch size is set to be 2 for training. The backbone networks for all experiments are pre-trained on the ImageNet dataset [34].

LLVIP: The input image is set to be 1280×1024 for both the training and test stages. We train the network for 12 epochs in total and reduce the learning rate at 8 and 11 epochs.

KAIST: The input image is set to be 640×512 . Our proposed method is trained for 6 epochs. For Faster RCNN and Cascade RCNN, the learning rate is set to be 2.5×10^{-3} for the first 4 epochs, and 2.5×10^{-4} for the last 2 epochs. In respect of RetinaNet, we start with the initial learning rate of 1.25×10^{-3} for the first 4 epochs and further decay it to 1.25×10^{-4} for the last 2 epochs.

4.3 Comparsion on LLVIP Dataset

We perform comprehensive ablation studies to validate the effectiveness of our proposed modules on the LLVIP *test* set.

Baseline comparison: First we show the impacts of our proposed modules: local dynamic cross-modal (LDCM) module and non-local dynamic cross-modal (NDCM) module on the LLVIP *test* set as given in Tab. 1. For a fair comparison, the detector and backbone network for all methods in Tab. 1 are RetinaNet and ResNet50, respectively. We utilize feature concatenation as the baseline feature aggregation manner. The baseline obtains 56.9% AP on the *test* set. Further, integrating the LDCM module leads to an improvement of 1.3%. The introduction of the NDCM module improves the performance from 56.9% to 58.3% AP. Our novel DCMNet (including the LDCM and the NDCM as whole) provides a gain of 2.0% over the baseline RetinaNet. The experimental results demonstrate the effectiveness of our proposed modules.

Impacts of the number of convolutional kernels in LDCM:

Here, we analyze the impacts of different numbers of convolutional kernels in LDCM, and the results are given in Tab. 2(left). We consider three numbers: 2, 4, and 8. For a fair comparison, except the number of convolutional kernels, other architectures and experimental settings are kept the same. We choose 4 as it provides an optimal trade-off between the detection accuracy and the speed.

Impacts of the number of latent spaces in NDCM: Here, we analyze the impacts of different numbers of latent spaces in NDCM, and results are given in Tab. 2 (right). We consider three numbers: 4, 8, and 16. For a fair comparison, except the number of latent spaces, other architectures and experimental settings are kept the

Table 2: The impacts of different numbers of convolutional kernels in LDCM and latent spaces in NDCM on the LLVIP *test* set. CK: convolutional kernels, LS: latent spaces.

# CK	2	4	8	# LS	4	8	16
AP	57.9	58.2	58.2	AP	57.9	58.3	58.3

Table 3: Comparison (in average precision (%)) of our method with different detectors and different feature fusion manners on the LLVIP *test* set. For a fair comparison, all the methods use the same backbone (ResNet50), training data, input scale and experimental settings.

Detectors		Fusion Methods	AP $\uparrow$
Two-stage	Faster RCNN	Baseline	56.9
		DCMNet	58.4($\uparrow$1.5)
	Cascade RCNN	Baseline	59.6
		DCMNet	61.5($\uparrow$1.9)
Single-stage	RetinaNet	Baseline	56.9
		DCMNet	58.9($\uparrow$2.0)
	SABL	Baseline	58.5
		DCMNet	60.6($\uparrow$2.1)
Anchor-free	Reppoints	Baseline	57.6
		DCMNet	58.7($\uparrow$1.1)

same. We choose 8 as the number of latent spaces as it provides an optimal trade-off between the detection accuracy and the speed.

Robustness on different detectors: Here, we integrate our DCMNet into different popular detectors: RetinaNet [21], SABL [37], Faster RCNN[32], Cascade RCNN [1], and RepPoints [44]. For a fair comparison, except for the detectors, all the methods utilize the same backbone (ResNet50) and experimental settings. The results are provided in Tab. 3. Compared with the baseline feature fusion manner (*i.e.*, concatenation), integrating our DCMNet into Faster RCNN, Cascade RCNN, RetinaNet, SABL, and Reppoints obtains 1.5%, 1.9%, 2.0%, 2.1%, and 1.1% AP improvements, respectively. Note that compared with the baseline, integrating our DCMNet into different detectors improves the detection performance of all detectors including two-stage, single-stage, and anchor-free detectors. The experimental results demonstrate the effectiveness and the *generality* of our DCMNet.

Robustness on different backbone networks: Here, we also evaluate effects of different backbone networks with our DCMNet in terms of average precision on the LLVIP *test* set. We conduct experiments on three different backbone networks: ResNet50 [9], ResNeSt50 [46], and PVT-v2 [38]. For a fair comparison, when changing the backbone, the entire detection architecture and parameter settings remain the same. We report results in Tab. 4. Compared with the baseline fusion method (*i.e.*, concatenation), our DCMNet obtains absolute gains of 2.0%, 2.4% and 1.4% on ResNet50, ResNeSt50 and PVT-v2, respectively. The results in Tab. 4 demonstrate that our DCMNet is robust on *various* backbone networks including CNN-based and Transformer-based networks.

Comparison with non-local networks [39]: Here, we conduct an experiment to compare our NDCM module and the non-local network. Non-local network is one of the most popular methods

¹<https://github.com/open-mmlab/mmdetection>

Table 4: Comparisons (in average precision (%)) of our method with different backbones and different feature fusion manners on the LLVIP *test* set. For a fair comparison, we choose the same detector (RetinaNet) as the baseline. All the methods use the same training data, input scale and experimental settings.

Backbones	Type	Fusion Methods	AP $\uparrow$
ResNet50	CNN	Baseline	56.9
		DCMNet	58.9($\uparrow$2.0)
ResNeSt50	CNN	Baseline	59.1
		DCMNet	61.5($\uparrow$2.4)
PVT-V2	Transformer	Baseline	59.3
		DCMNet	60.7($\uparrow$1.4)

Table 5: Comparison (in average precision (%) and running speed (s)) with different feature aggregation methods on the LLVIP *test* set. All methods adopt the same detector and backbone (i.e., RetinaNet and ResNet). The size of input image is 1280×1024 . *L*: Local manner, *NL*: Non-local manner.

Methods	Type	AP $\uparrow$	Running Speed
Concatenation	<i>L</i>	56.9	0.095
NL-Net [39]	<i>NL</i>	57.3	0.135
NDCM (ours)	<i>NL</i>	58.3	0.099
DCMNet (Ours)	<i>L + NL</i>	58.9	0.107

to model the long-range dependencies. We employ the non-local network after both of the RGB features and infrared features from the backbone network to model long-range dependencies of intra-modalities, and then aggregate these features. The results are provided in Tab. 5. For a fair comparison, all methods in Tab. 5 employ the same detector, backbone, and experimental settings, respectively. It can be observed that compared with the way of employing non-local network, our NDCM module obtains superior detection accuracy with faster speed. We summarize the following two reasons : (i) The non-local network only captures the long-range dependencies of intra-modalities. On the contrary, our NDCM module models long-range dependencies of cross-modalities. The long-range dependencies of cross-modalities help in improving the detection accuracy by exploring the non-local complementary information between the two modalities. (ii) Because the number of the latent feature nodes (in our methods, we set it to be 128) is much smaller than the number of pixels (the size of the input image is 1280×1024 , the size of the features from ResNet Block3 is 160×128 , the number of pixels is 20480), our method is faster than the non-local network.

Computational complexity analysis: Here, we analyze the computational complexity of our DCMNet. Tab. 5 shows the running time of our DCMNet and baseline (feature concatenation). The results in the Tab. 5 is measured on a single NVIDIA RTX3090 GPU. For a fair comparison, both our DCMNet and the baseline are integrated into the RetinaNet with ResNet50. When the input resolution is 1280×1024 , compared with the baseline, our DCMNet only has additional 0.012s for one image on the LLVIP *test* set. Although the test time of our DCMNet is slightly slower than the baseline,



Figure 5: Feature visualizations. First row: the RGB images and features. Second row: the infrared images and features. First and fourth column: the input images. Second and fifth column: the input of the LDCM. Third and sixth column: the output of the LDCM.

Table 6: Comparison (in log-averaged miss rate (%)) of our DCMNet with the baseline feature fusion manner on the KAIST *test* set. For a fair comparison, all the methods use the same backbone network (VGG16) and experimental settings.

Detectors	Fusion Methods	$MR^{-2} \downarrow$
RetinaNet	Baseline	9.02
	DCMNet	6.89($\downarrow$2.13)
Faster RCNN	Baseline	8.27
	DCMNet	6.41($\downarrow$1.86)
Cascade RCNN	Baseline	7.91
	DCMNet	6.17($\downarrow$1.74)

our DCMNet significantly improves the detection accuracy from 56.9% to 58.9%. Therefore, it can be concluded that the proposed DCMNet results in better detection accuracy at the cost of a slight computational burden.

Qualitative comparison: Fig. 5 shows the feature maps of the input and output of our proposed LDCM module. As shown in Fig. 5, the input feature maps contain irrelevant information. We observe that in contrast to the input, the output of our LDCM module helps in better discriminating object regions from the background thereby leading to an improvement of the detection accuracy.

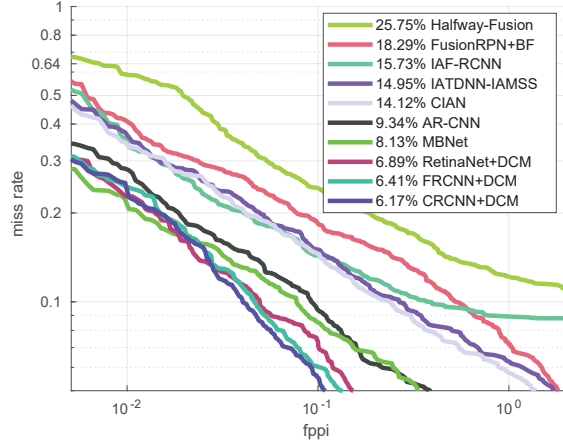
4.4 Comparison on the KAIST Dataset

Here, we evaluate the performance on the KAIST dataset. We compare our proposed DCMNet with the baseline (feature concatenation) on different detectors including RetinaNet [21], Faster RCNN [32], and Cascade RCNN [1] on the KAIST *test* set. For a fair comparison, both of the baseline and our DCMNet utilize the same backbone (VGG16) and other experimental settings. The results are provided in Tab. 6. Compared with the baseline, integrating our DCMNet into RetinaNet, Faster RCNN, and Cascade RCNN reduces the log-averaged miss rates by 2.13%, 1.86%, and 1.74%, respectively. Note that compared with the baseline, our DCMNet reduces the log-averaged miss rate on different detectors on the KAIST *test* set.

State-of-the-art comparison: Here, our DCMNet is compared with the following recent state-of-the-art methods: Halfway Fusion [23], Fusion RPN [15], Fusion RPN+BF [15], IAF-RCNN [17], CIAN [49], MSDS-RCNN [16], AR R-CNN [51], MBNet [59], and BAANet [43]. Tab. 7 gives the results on the KAIST *test* set. Among existing methods, BAANet [43] provides the best result with a log-average miss rate of 7.92%, and our DCMNet with Cascade RCNN outperforms BAANet with an absolute gain of 1.75%. Fig. 6 shows the comparison of our method with existing methods over the whole

Table 7: Comparison with the state-of-the-art methods in terms of log-averaged miss-rate on the KAIST *test* set.

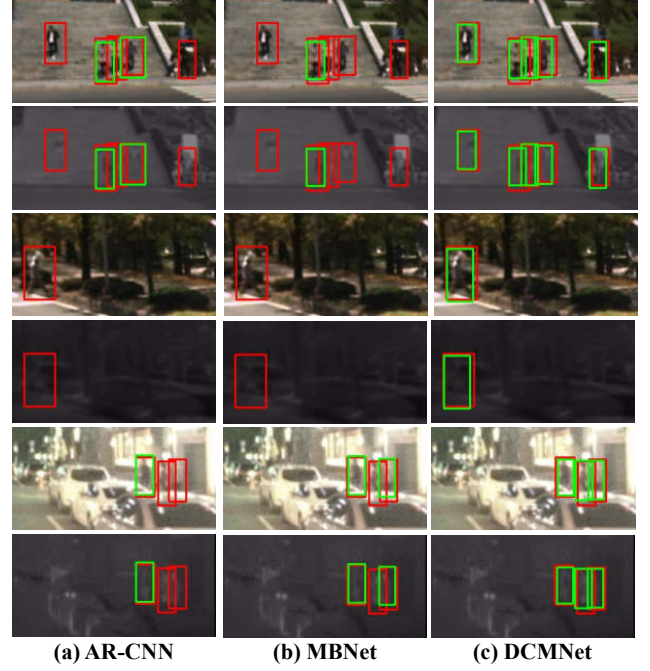
Methods	Backbone	$MR^{-2} \downarrow$
Halfway Fusion [23]	VGG16	25.75
Fusion RPN+BF [15]	VGG16	18.29
IAF-RCNN [17]	VGG16	15.73
IATDNN+IAMSS [7]	VGG16	14.95
CIAN [49]	VGG16	14.12
MSDS-RCNN [16]	VGG16	11.63
AR-CNN[51]	VGG16	9.34
MBNet [59]	ResNet50	8.13
BAANet [43]	ResNet50	7.92
RetinaNet + DCMNet	VGG16	6.89
Faster RCNN + DCMNet	VGG16	6.41
Cascade RCNN + DCMNet	VGG16	6.17

**Figure 6: Comparison on the the KAIST *test* set. The legend in each plot represents the log-averaged miss rate. Our approach provides superior results compared to existing approaches.**

spectrum of false positives per image (FPPI) metric. We further signify the effectiveness of DCMNet towards detecting pedestrians under various illuminations by drawing visual comparisons with MBNet [59] and AR-CNN [51] in Fig. 7. We choose Faster RCNN as the detector of our DCMNet. For a fair comparison, all results are obtained using the same FPPI (false positive per image), which is a common practice utilized in [17, 30]. Our method accurately detects pedestrians in all three scenarios.

Our DCMNet achieves the state-of-the-art performance. Compared with exiting methods which usually use element-wise operations to fuse features across different modalities, our DCMNet has two advantages: (i) Our DCMNet fuses features by adaptively utilizing the complementary information between the two modalities. It is likely to reduce the noise introduced by element-wise operations. (ii) Our DCMnet can capture the long-range dependencies across different modalities which are beneficial to reason the existence and location of the pedestrians.

Different training settings: Some state-of-the-art methods [16, 45] are trained on the sanitized annotations of KAIST which filter the annotation noise. For a fair comparison, we also train our method on the sanitized annotations. The results are provided

**Figure 7: Qualitative comparison of (c) DCMNet with (a) AR-CNN [51] and (b) MBNet [59] on the KAIST *test* set. Red boxes denote ground-truth and green boxes indicate detector predictions. Example images depict varying illuminations.****Table 8: Comparison with the state-of-the-art methods on the KAIST *test* set. The training set is the sanitized annotations provided by [16].**

Methods	Backbone	$MR^{-2} \downarrow$
MSDS-RCNN [16]	VGG16	7.49
GAFF [45]	VGG16	6.48
Faster RCNN+DCMNet	VGG16	5.84

in Tab. 8. It can be observed that our method with Faster RCNN achieves new state-of-the-art performance.

5 CONCLUSION

We proposed a dynamic cross-modal network (DCMNet) for multispectral pedestrian detection. It reduces the local irrelevant and distracting information during multispectral feature fusion through a local dynamic cross-modal (LDCM) module and models the cross-modality long-range dependencies to improve the detection accuracy through a non-local dynamic cross-modal (NDCM) module. Our DCMNet is applied to a variety of detectors (including single-stage, two stage, and anchor-free detectors) and backbone networks (including CNN-based and Transformer-based networks), and achieves consistent improvements. Extensive experiments on two challenging multispectral pedestrian detection benchmarks clearly show the effectiveness of our approach.

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